## Aim:

Implement MongoDB aggregation to summarize student grades, with schema design.

## Description:

This program uses MongoDB aggregation to analyze student subject marks and generate a grade summary. It calculates total, average, highest, and lowest marks for each student using a structured student schema.

## Algorithm:

- Design the MongoDB schema to store student ID, name, department, and subject marks.
- Connect to the MongoDB database and select the students collection.
- Use $unwind to process each subject record separately.
- Use $group to calculate total, average, highest, and lowest marks for each student.
- Use $project and $sort to format and display the final grade summary.

## Program:

**Data connection & Data insertion**

```
const { MongoClient } = require("mongodb");
const url = "mongodb://localhost:27017";
const client = new MongoClient(url);
const dbName = "studentDB";
async function main() {
  try {
    await client.connect();
    console.log("Connected to MongoDB successfully!");
    const db = client.db(dbName);
    const collection = db.collection("students");
  } catch (error) {
    console.error("Error connecting to MongoDB:", error);
  }
}
main();
```

**Inserting Student Records into MongoDB Collection**

```
const { MongoClient } = require("mongodb");
const url = "mongodb://localhost:27017";
const client = new MongoClient(url);
const dbName = "studentDB";
async function main() {
  try {
    await client.connect();
    console.log("Connected to MongoDB successfully!");
    const db = client.db(dbName);
    const collection = db.collection("students");
    const students = [
      {
        studentId: 101,
        name: "Arjun",
        department: "CSE",
        subjects: [
          { subject: "DBMS", marks: 85, grade: "A" },
          { subject: "AI", marks: 90, grade: "A+" },
          { subject: "Web Development", marks: 78, grade: "B+" }
        ]
      },
      {
        studentId: 102,
        name: "Rohan",
        department: "CSE",
        subjects: [
          { subject: "DBMS", marks: 75, grade: "B+" },
          { subject: "AI", marks: 82, grade: "A" },
```

```
      { subject: "Web Development", marks: 88, grade: "A" }
    ]
  },
  {
    studentId: 103,
    name: "Sneha",
    department: "AIML",
    subjects: [
      { subject: "DBMS", marks: 92, grade: "A+" },
      { subject: "AI", marks: 95, grade: "A+" },
      { subject: "Web Development", marks: 89, grade: "A" }
    ]
  },
  {
    studentId: 104,
    name: "Varun",
    department: "AIML",
    subjects: [
      { subject: "DBMS", marks: 68, grade: "B" },
      { subject: "AI", marks: 74, grade: "B+" },
      { subject: "Web Development", marks: 80, grade: "A" }
    ]
  }
];
await collection.deleteMany({});
await collection.insertMany(students);
console.log("Student records inserted successfully!");
} catch (error) {
console.error(error);
```

```
  } finally {
    await client.close();
  }
}
main();
```

**Complete Grade Summary Aggregation**

```
const { MongoClient } = require("mongodb");
const url = "mongodb://localhost:27017";
const client = new MongoClient(url);
const dbName = "studentDB";
async function main() {
    try {
        await client.connect();
        console.log("Connected to MongoDB successfully!");
        const db = client.db(dbName);
        const collection = db.collection("students");
        const students = [
            {
                studentId: 101,
                name: "Arjun",
                department: "CSE",
                subjects: [
                    { subject: "DBMS", marks: 85, grade: "A" },
                    { subject: "AI", marks: 90, grade: "A+" },
                    { subject: "Web Development", marks: 78, grade: "B+" }
                ]
            },
            {
                studentId: 102,
```

```
    name: "Rohan",
    department: "CSE",
    subjects: [
      { subject: "DBMS", marks: 75, grade: "B+" },
      { subject: "AI", marks: 82, grade: "A" },
      { subject: "Web Development", marks: 88, grade: "A" }
    ]
  },
  {
    studentId: 103,
    name: "Sneha",
    department: "AIML",
    subjects: [
      { subject: "DBMS", marks: 92, grade: "A+" },
      { subject: "AI", marks: 95, grade: "A+" },
      { subject: "Web Development", marks: 89, grade: "A" }
    ]
  },
  {
    studentId: 104,
    name: "Varun",
    department: "AIML",
    subjects: [
      { subject: "DBMS", marks: 68, grade: "B" },
      { subject: "AI", marks: 74, grade: "B+" },
      { subject: "Web Development", marks: 80, grade: "A" }
    ]
  }
];
```

```
await collection.deleteMany({});
await collection.insertMany(students);
console.log("Student records inserted successfully!");
const result = await collection.aggregate([
  {
    $unwind: "$subjects"
  },
  {
    $group: {
      _id: "$studentId",
      name: {
        $first: "$name"
      },
      department: {
        $first: "$department"
      },
      totalMarks: {
        $sum: "$subjects.marks"
      },
      highestMarks: {
        $max: "$subjects.marks"
      },
      lowestMarks: {
        $min: "$subjects.marks"
      },
      averageMarks: {
        $avg: "$subjects.marks"
      }
    }
```

```
    },
    {
      $project: {
        _id: 0,
        name: 1,
        department: 1,
        totalMarks: 1,
        highestMarks: 1,
        lowestMarks: 1,
        studentId: "$_id",
        averageMarks: {
          $round: ["$averageMarks", 2]
        }
      }
    },
    {
      $sort: {
        averageMarks: -1
      }
    }
]).toArray();
const finalResult = result.map((student, index) => ({
  index: index + 1,
  name: student.name,
  department: student.department,
  totalMarks: student.totalMarks,
  highestMarks: student.highestMarks,
  lowestMarks: student.lowestMarks,
  studentId: student.studentId,
```

```
            averageMarks: student.averageMarks
        }));
        console.log("\nStudent Grade Summary:");
        console.table(finalResult);
    } catch (error) {
        console.error("Error:", error);
    } finally {
        await client.close();
    }
}
main();
```

```
PS D:\student-grade-mongodb> node app.js
Connected to MongoDB successfully!
PS D:\student-grade-mongodb> node app.js
Connected to MongoDB successfully!
Student records inserted successfully!
PS D:\student-grade-mongodb> node app.js
Connected to MongoDB successfully!
Student records inserted successfully!

Student Grade Summary:
```

| (index) | name | studentId | averageMarks |
|---|---|---|---|
| 0 | 'Sneha' | 103 | 92 |
| 1 | 'Arjun' | 101 | 84.33 |
| 2 | 'Rohan' | 102 | 81.67 |
| 3 | 'Varun' | 104 | 74 |

```
PS D:\student-grade-mongodb> node app.js
Connected to MongoDB successfully!
Student records inserted successfully!

Student Grade Summary:
```

| (index) | index | name | department | totalMarks | highestMarks | lowestMarks | studentId | averageMarks |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 'Sneha' | 'AIML' | 276 | 95 | 89 | 103 | 92 |
| 1 | 2 | 'Arjun' | 'CSE' | 253 | 90 | 78 | 101 | 84.33 |
| 2 | 3 | 'Rohan' | 'CSE' | 245 | 88 | 75 | 102 | 81.67 |
| 3 | 4 | 'Varun' | 'AIML' | 222 | 80 | 68 | 104 | 74 |

```
PS D:\student-grade-mongodb>
```

